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4. (a) Using a suitable signaling pathways, illustrate in detail the process of intracellular signal transduction and its role in target gene regulation. (12)
- (b) Briefly describe the concept of secondary messenger in cell to cell signaling. (3)
5. (a) Explain in detail the hallmarks of cancer. Give a brief explanation of its development process. (8)
- (b) Describe the role of viruses in development of cancer either in experimental animals or humans. (7)
6. Describe in detail the molecular process of transcription regulation in eukaryotes with appropriate examples (15)
7. Write short notes on any **three** of the following : (5,5,5)
- Signal molecules
  - Molecular approaches to cancer treatment
  - Stages of meiosis
  - Role of SMCs in transcription regulation

(1000)

[This question paper contains 4 printed pages.]

Your Roll No. ....

Sr. No. of Question Paper : 4259 H

Unique Paper Code : 2233012006

Name of the Paper : DSE-4, Cell Growth and Regulation

Name of the Course : B.Sc. (H) Zoology, (NEP)

Semester : IV (DSE-4)

Duration : 3 Hours Maximum Marks : 90

**Instructions for Candidates**

- Write your Roll No. on the top immediately on receipt of this question paper.
- Attempt any **five** questions in all including question No. 1, which is compulsory.
- Illustrate your answers with appropriate well-labelled diagrams whenever necessary.

- (a) Define the following (**any six**) : (1x6)
  - Pluripotency
  - Transdifferentiation
  - Cyclins
  - Necrosis
  - Metastasis

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(vi) Transcription repressor

(vii) Cytokinesis

(b) Differentiate between the following (**any four**) :  
(2x4)

(i) Condensins and cohesins

(ii) Intrinsic and extrinsic pathways of apoptosis

(iii) Lymphoma and leukemia

(iv) Adult stem cells and embryonic stem cells

(v) Autocrine and paracrine signaling

(c) Write the significance of the following (**any four**) :  
(1x4)

(i) Spindle assembly check point

(ii) cAMP phosphodiesterase

(iii) p53

(iv) Enhancers

(v) Ras protein

(d) Name the following :

(i) A signal molecule from plant

(ii) A retrovirus that can cause human tumors

(iii) Surface receptor in extrinsic pathways of apoptosis

(iv) A human cell in  $G_0$  state. (1x4)

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(e) Fill in the blanks : (1x4)

(i) \_\_\_\_\_ are the example of cis-acting sequences.

(ii) \_\_\_\_\_ are malignancies of epithelial cells.

(iii) \_\_\_\_\_ are heterotrimeric proteins and are coupled to GPCR.

(iv) \_\_\_\_\_ complex is involved in G1 to S transition.

(f) Expand the following (**any four**) :- (1x4)

(i) MOMP

(ii) CKI

(iii) MCC

(iv) HDACs

(v) PDGF

2. Discuss in details the role of CDKs and cyclins in various phases of cell cycle. What are the mechanisms that regulate CDKs. (15)

3. (a) Explain in detail the events of programmed cell death including the role of proteins that regulate apoptosis. (12)

(b) Tabulate the applications of stem cells in regenerative medicine. (3)

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